

## 0.1 Section 1.3 - Logical Equivalence

### Recall 1 (Logical Equivalence)

Recall that two propositions are **logically equivalent** if they have the same truth table

### 0.1.1 Testing for Equivalence using Truth Tables

#### Example 1

Determine if the following are logically equivalent

1.  $p \implies q$  and  $\neg p \vee q$

*Solution:*

| $p$ | $q$ | $p \implies q$ | $\neg p$ | $\neg p \vee q$ |
|-----|-----|----------------|----------|-----------------|
| T   | T   | T              | F        | T               |
| T   | F   | F              | F        | F               |
| F   | T   | T              | T        | T               |
| F   | F   | T              | T        | T               |

$$\therefore p \implies q \equiv \neg p \vee q$$

2.  $\neg(p \wedge q)$  and  $\neg p \vee \neg q$

*Solution:*

| $p$ | $q$ | $p \wedge q$ | $\neg(p \wedge q)$ | $\neg p$ | $\neg q$ | $\neg p \vee \neg q$ |
|-----|-----|--------------|--------------------|----------|----------|----------------------|
| T   | T   | T            | F                  | F        | F        | F                    |
| T   | F   | F            | T                  | F        | T        | T                    |
| F   | T   | F            | T                  | T        | F        | T                    |
| F   | F   | F            | T                  | T        | T        | T                    |

$$\therefore \neg(p \wedge q) \equiv \neg p \vee \neg q$$

3.  $(p \implies q) \vee (q \implies r)$  and  $p \implies (q \vee r)$

*Solution:*

| $p$ | $q$ | $r$ | $p \implies r$ | $q \implies r$ | $(p \implies r) \vee (q \implies r)$ | $q \vee r$ | $p \implies (q \vee r)$ |
|-----|-----|-----|----------------|----------------|--------------------------------------|------------|-------------------------|
| T   | T   | T   | T              | T              | T                                    | T          | T                       |
| T   | T   | F   | F              | F              | F                                    | T          | T                       |
| T   | F   | T   | T              | T              | T                                    | T          | T                       |
| T   | F   | F   | F              | T              | T                                    | F          | F                       |
| F   | T   | T   | T              | T              | T                                    | T          | T                       |
| F   | T   | F   | T              | F              | T                                    | T          | T                       |
| F   | F   | T   | T              | T              | T                                    | T          | T                       |
| F   | F   | F   | T              | T              | T                                    | F          | T                       |

$$\therefore (p \implies q) \vee (q \implies r) \not\equiv p \implies (q \vee r)$$

So far we know the following:

- $p \implies q \equiv \neg p \vee q$
- DeMorgan's Law
  - $\neg(p \vee q) \equiv \neg p \wedge \neg q$
  - $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- Commutative (swap places)
  - $p \vee q \equiv q \vee p$
  - $p \wedge q \equiv q \wedge p$
- Associative (swap grouping)
  - $(p \vee q) \vee r \equiv p \vee (q \vee r)$
  - $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- Distributive (distributing operations)
  - $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
  - $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- Identity
  - $p \wedge T \equiv p$
  - $p \vee F \equiv p$
- Domination Laws
  - $p \vee T \equiv T$
  - $p \wedge F \equiv F$
- Impotent Laws

$$- p \vee p \equiv p$$

$$- p \wedge p \equiv p$$

### **Definition 1 (Tautology and Contradiction)**

A proposition that is always **TRUE** is a **tautology**

A proposition that is always **FALSE** is a **contradiction**

### **Note**

When proving tautologies or contradictions, you could either:

1. Use Truth Tables
2. Use the Laws from earlier to reduce a compound proposition into **TRUE** or **FALSE**

But it's advised to use Truth Tables, since it's easy to get lost in your work when using the Laws

Also, ALWAYS start by **converting conditionals**

### **Example 2 (Examples of Tautologies and Contradictions)**

Determine if the following are tautologies, contradictions, or neither

$$1. (p \wedge q) \implies q$$

Proving with Truth Tables

| $p$ | $q$ | $p \wedge q$ | $q$ | $(p \wedge q) \implies q$ |
|-----|-----|--------------|-----|---------------------------|
| T   | T   | T            | T   | T                         |
| T   | F   | F            | F   | T                         |
| F   | T   | F            | T   | T                         |
| F   | F   | F            | F   | T                         |

$$\therefore (p \wedge q) \implies q \equiv \text{tautology}$$

Proving with Logical Equations

$$\begin{aligned}
 (p \wedge q) \implies q &\equiv \neg(p \wedge q) \vee q && \text{(Convert Conditional)} \\
 &\equiv (\neg p \vee \neg q) \vee q && \text{(DeMorgan's Law)} \\
 &\equiv \neg p \vee (\neg q \vee q) && \text{(Associative)} \\
 &\equiv \neg p \vee T \\
 &\equiv T && \text{(Domination)}
 \end{aligned}$$

$$\therefore (p \wedge q) \implies q \equiv \text{tautology}$$

2. There's another example but I'm too lazy

**Note**

When proving the **logical equivalence** of two propositions, start with the **MORE COMPLICATED** proposition

**Example 3 (Proving Logical Equivalence of Two Propositions)**

Determine if  $(p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$

$(p \implies r) \wedge (q \implies r)$  seems more complicated, so we'll start with that

$$\begin{aligned}(p \implies r) \wedge (q \implies r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{(Convert Conditionals)} \\ &\equiv (\neg p \wedge \neg q) \vee r && \text{(Un-Distribute)} \\ &\equiv \neg(p \vee q) \vee r && \text{(DeMorgan's Law)} \\ &\equiv (p \vee q) \implies r && \text{(Convert back to Conditional)}\end{aligned}$$

$$\therefore (p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$$